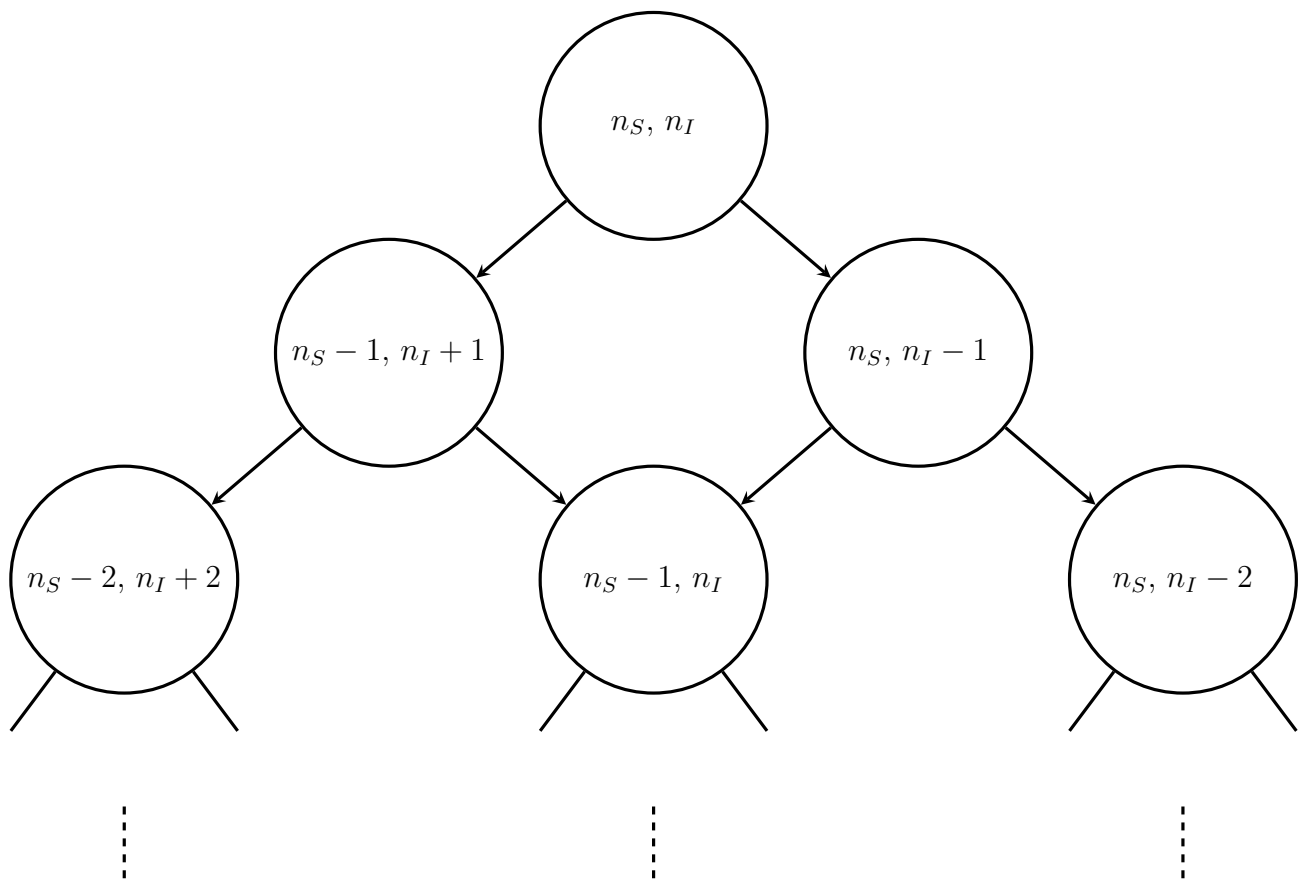
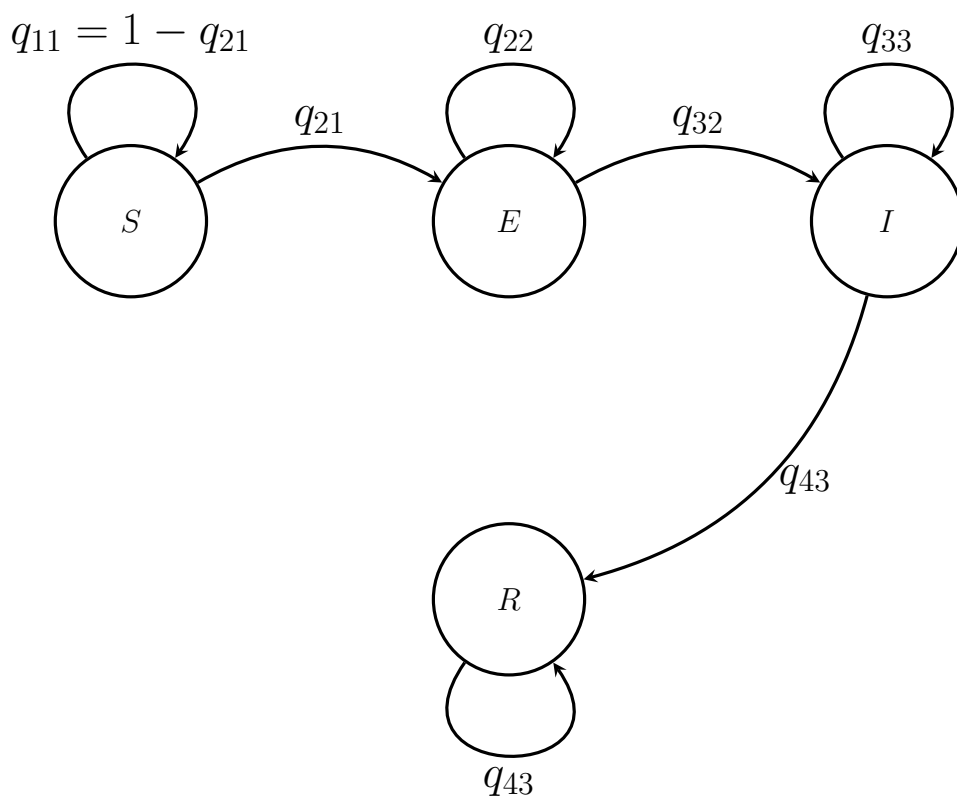
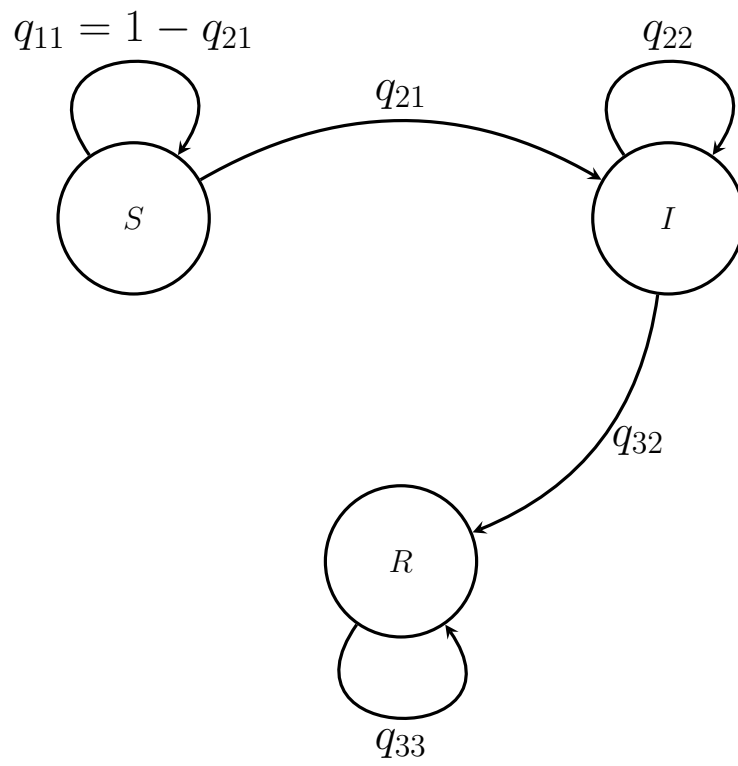
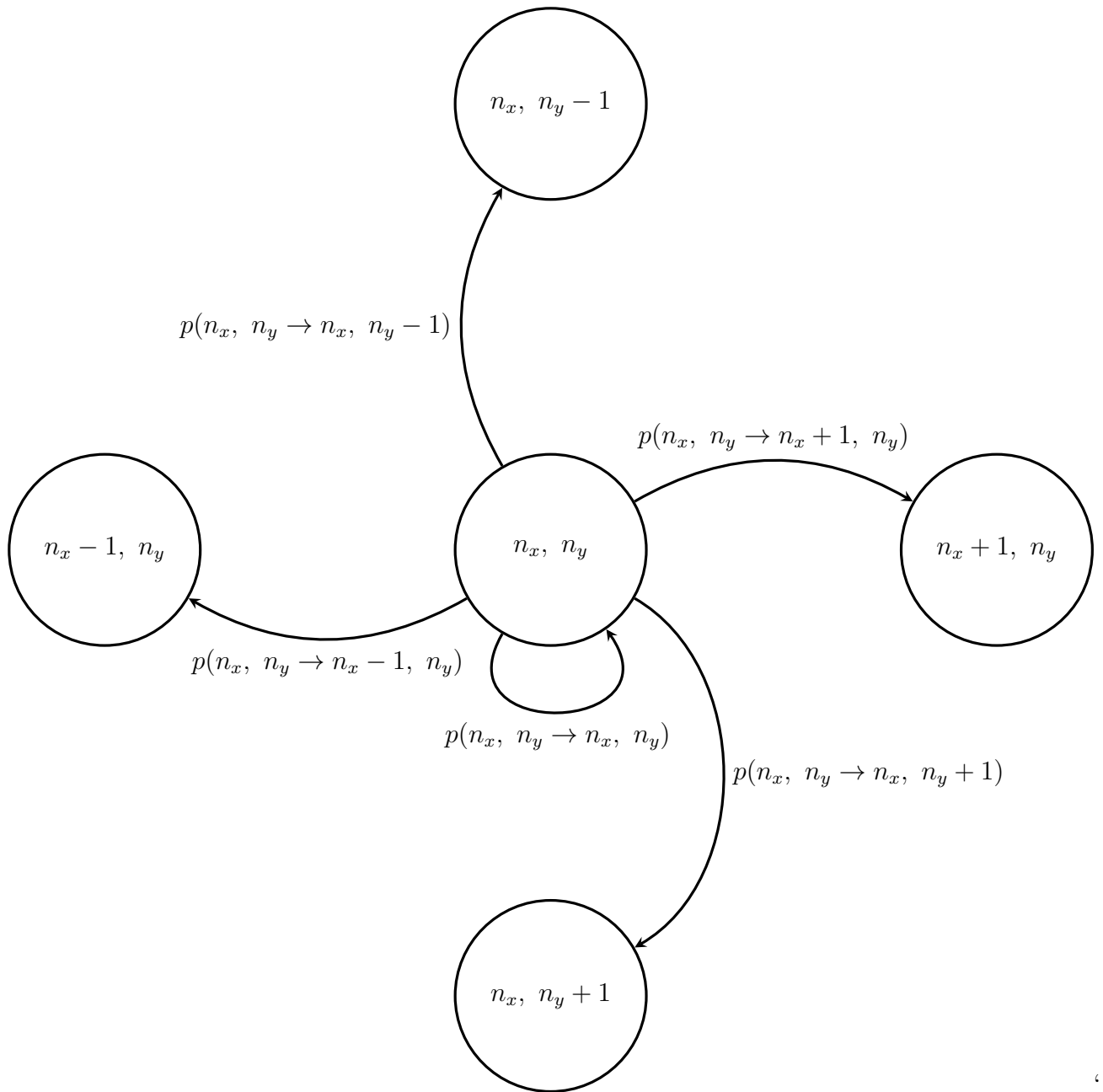


1 Markov Chain Diagrams







2 Plots

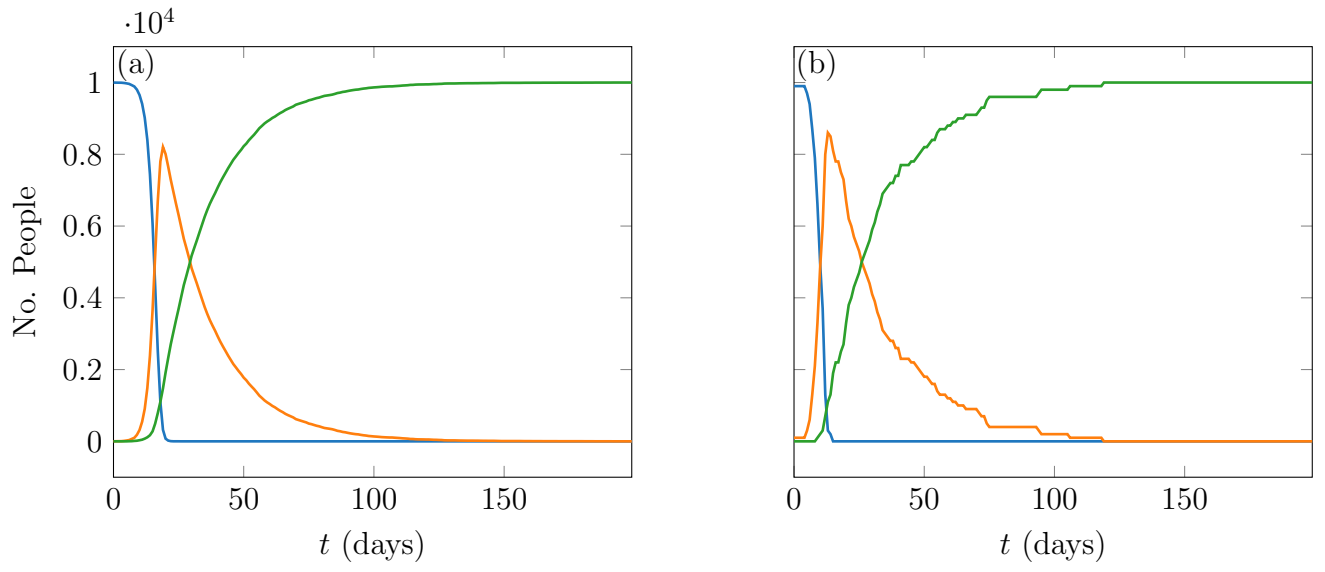


Figure 1: Microscopic SIR model with 100,000 people. System starts from 1 infection and 99,999 susceptible.

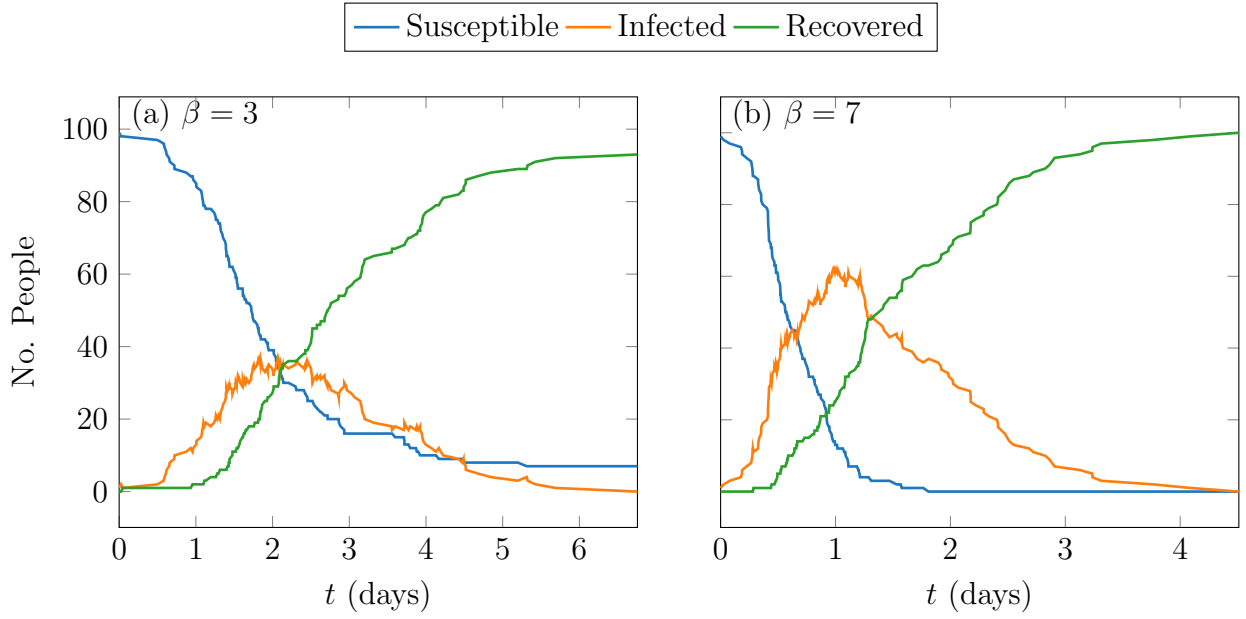


Figure 2: SIR model with β/N people infected per day per infected person per susceptible person and γ people recovered per day per infected person, where N is the total population. (a): $\beta = 3$, $\gamma = 1$; (b): $\beta = 7$, $\gamma = 1$. Figures produced by M. G. Hauge Evans.

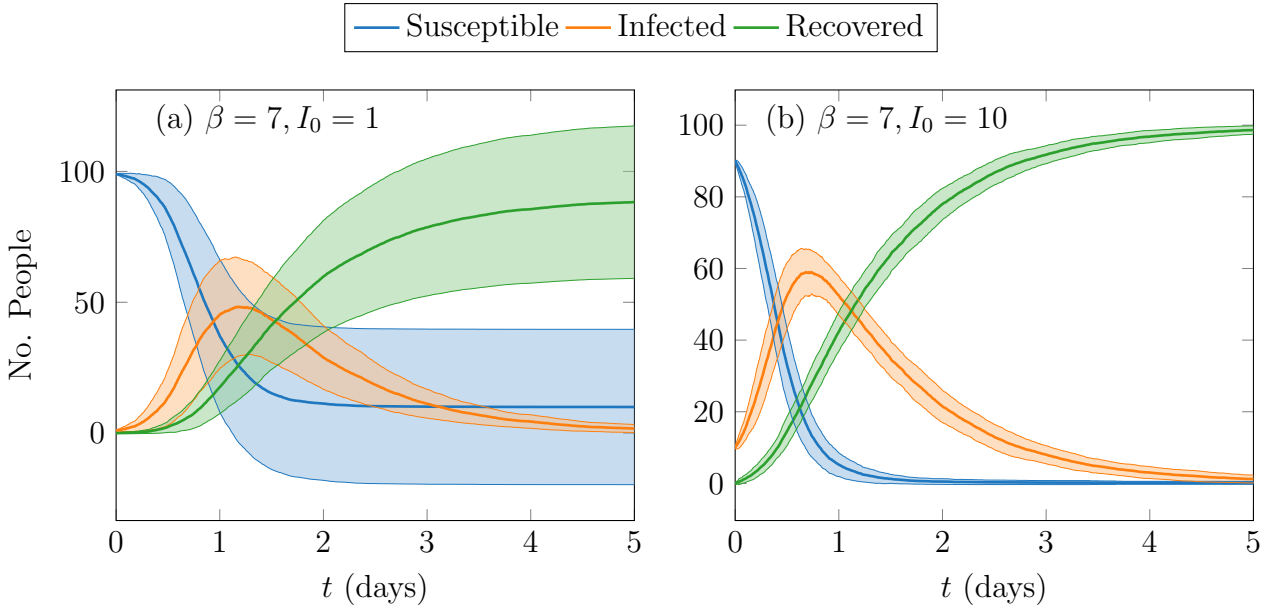


Figure 3: Ensemble averaged SIR model with β/N people infected per day per infected person per susceptible person and $\gamma = 1$ people recovered per day per infected person, where $N = 100$ is the total population. I_0 people are infected at $t = 0$ and the rest of the population is susceptible. (a): $\beta = 7$, $I_0 = 1$; (b): $\beta = 7$, $I_0 = 10$. Figures produced by M. G. Hauge Evans.

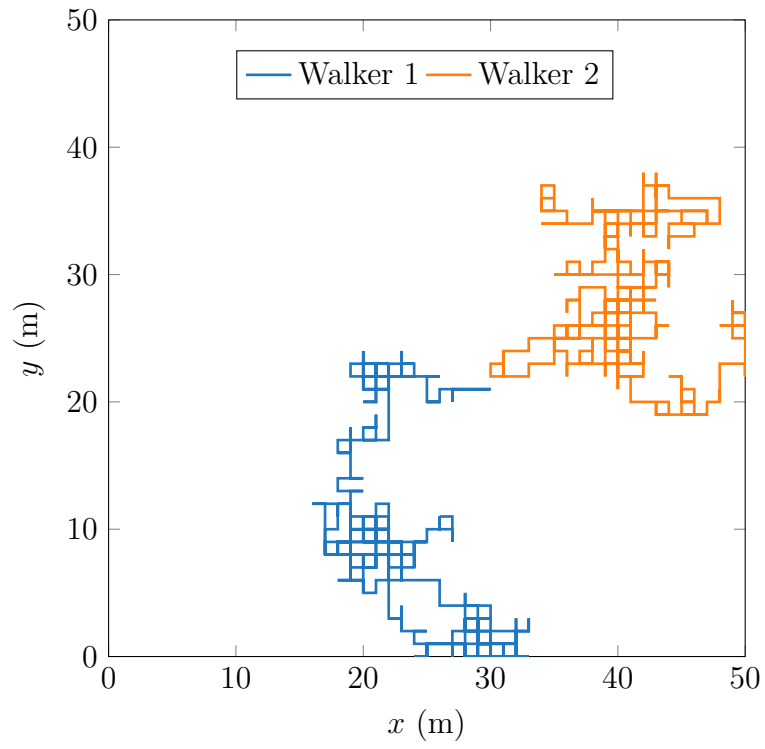
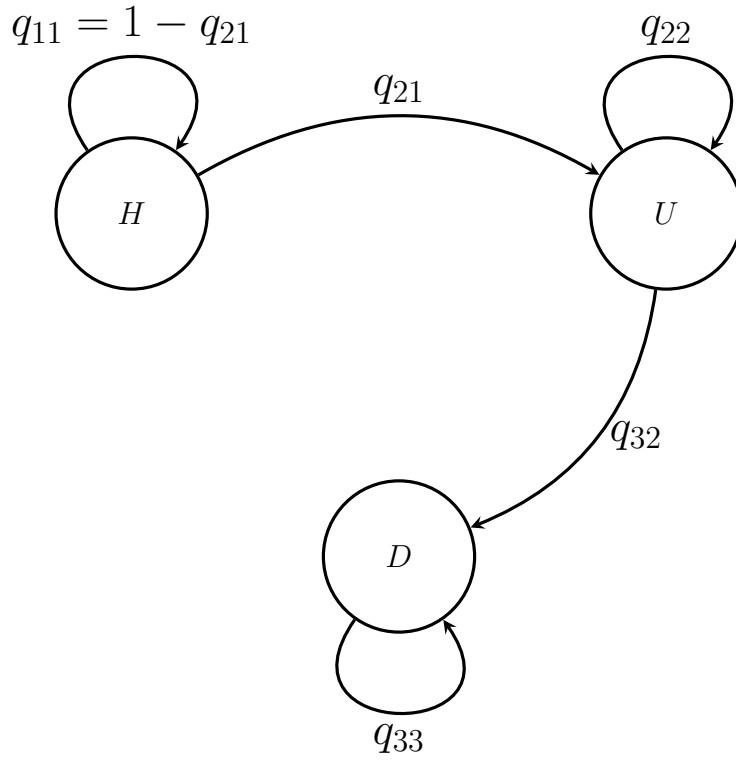


Figure 4: Random Walk Plot. Figures produced by M. G. Hauge Evans.



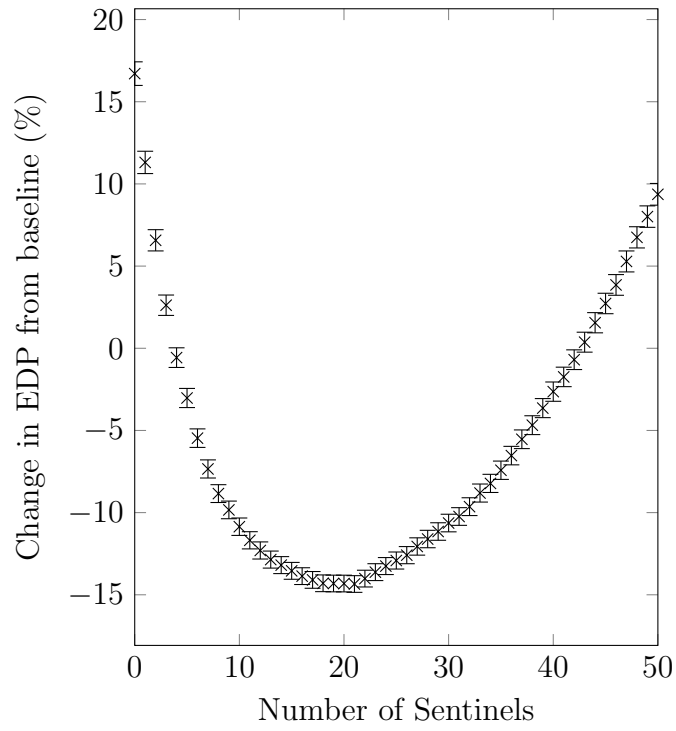


Figure 5

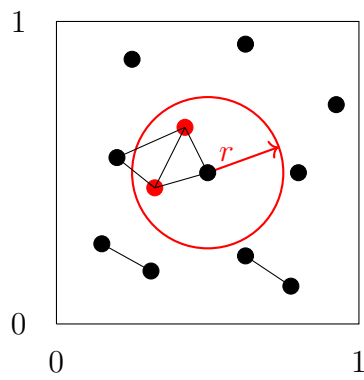


Figure 6: Figure depicting plants randomly plotted in a 1 x 1 area. Plants within some radius, r , of each other are linked on a network. Linked plants can spread the infection to one another.

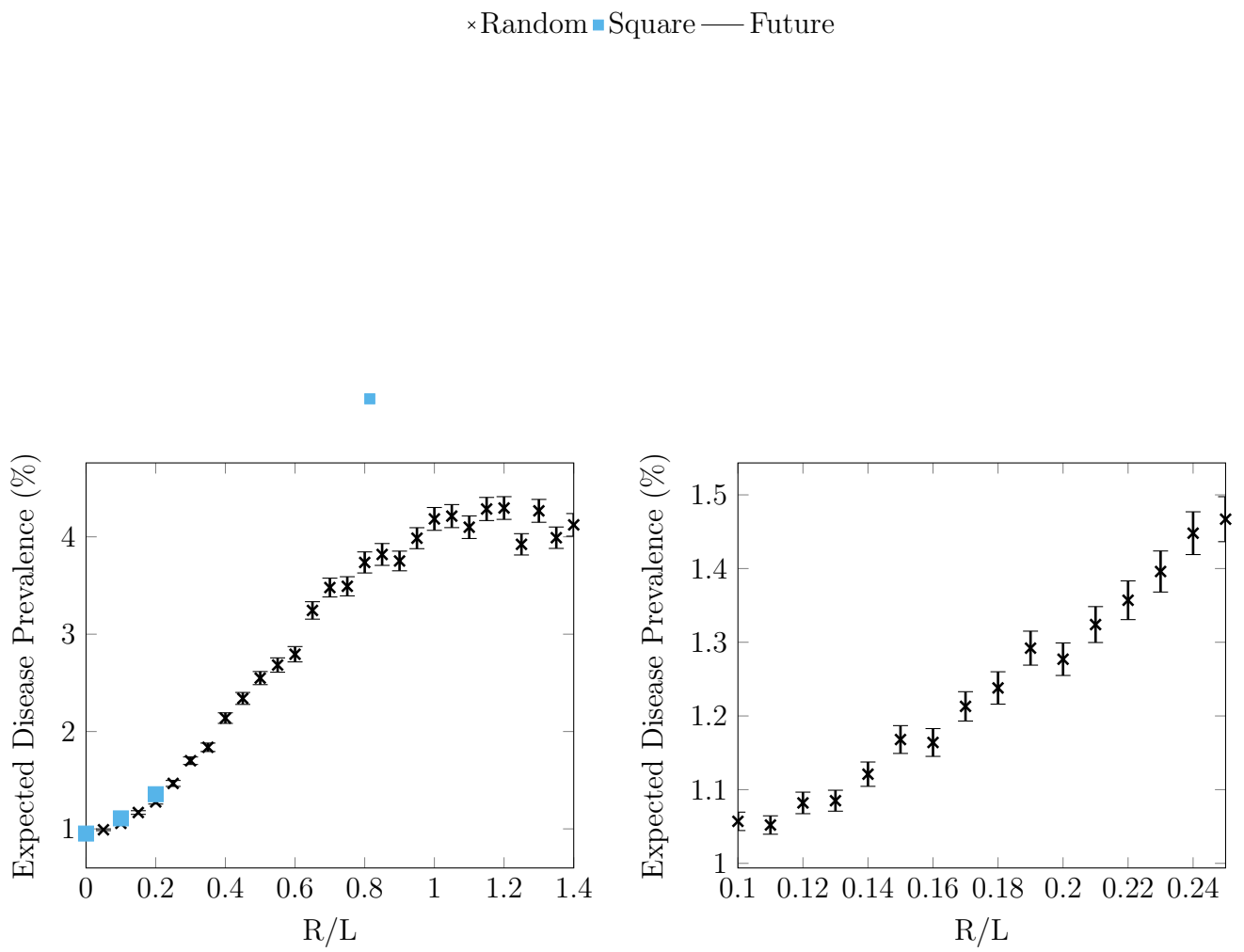


Figure 7: Microscopic SIR model with 100,000 people. System starts from 1 infection and 99,999 susceptible.